HVAC Controller System

Developed At

Aartronix Innovations Pvt. Ltd.

Role

Embedded Systems & Firmware Development

Technologies

STM32 • ESP32 • RS-485 • Modbus • Embedded C • HVAC Automation

Project Domain

Smart Building Automation & Industrial HVAC Control

Project Highlights

- Industrial-grade embedded HVAC controller
- Multi-board modular architecture
- Real-time STM32 firmware implementation
- RS-485 industrial communication
- ESP32 Wi-Fi monitoring integration
- Safe AC motor damper control
- Scalable zone-wise airflow management
- Future-ready IoT expansion support

HVAC Controller System

Overview

The HVAC Controller System is a modular embedded solution designed for smart building automation and efficient zone-wise airflow control. The system is developed using an STM32 microcontroller with industrial-grade RS-485 communication and ESP32 Wi-Fi integration for reliable monitoring and future IoT expansion.

The architecture is divided into three independent hardware modules:

- Controller Board
- Peripheral Board
- AC Motor Board

This modular approach improves scalability, maintenance, fault isolation, and real-time performance.

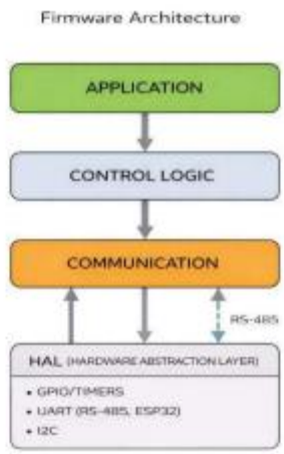
Key Features

- STM32-based real-time embedded control
- RS-485 communication for industrial reliability
- ESP32 Wi-Fi monitoring support
- Modular hardware architecture
- Relay-based AC motor damper control
- Limit switch protection for safe operation
- Multi-stage regulated power supply design
- Scalable multi-zone HVAC control
- Embedded firmware with layered architecture
- Future-ready IoT expansion support

System Architecture

1. Controller Board

The Controller Board acts as the central processing unit of the HVAC system.



Functions

- Executes control logic
- Handles RS-485 communication
- Interfaces with ESP32 Wi-Fi module
- Controls relay outputs
- Monitors limit switches
- Manages fault conditions

Main Components

- STM32 Microcontroller
- RS-485 Transceiver
- Relay Driver Circuits
- ESP32 Wi-Fi Module
- Power Regulation Circuits

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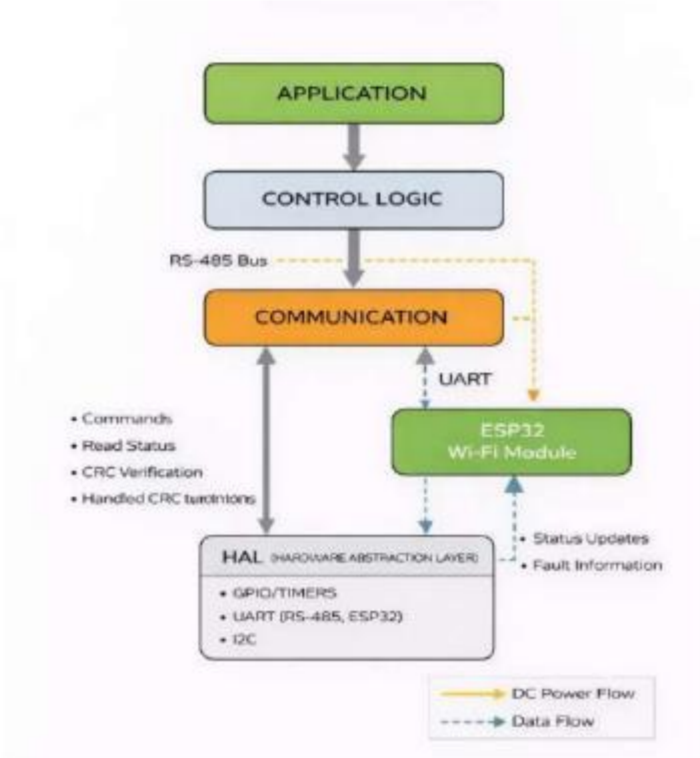
2. Peripheral Board

The Peripheral Board provides local user interaction and system diagnostics.



Features

- Pushbutton inputs
- LED status indicators
- RS-485 slave communication
- Local maintenance interface

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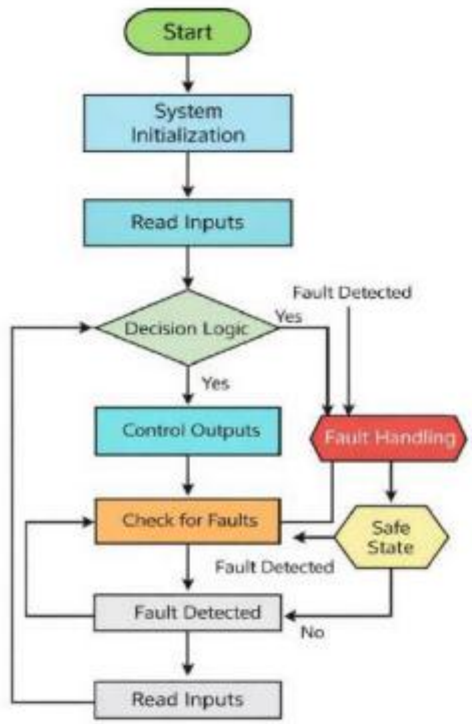
3. AC Motor Board

The AC Motor Board controls HVAC damper motors.



Features

- 24 VAC motor control
- Relay-driven actuation
- Normally Closed (NC) limit switches
- Safe damper positioning
- Zone connector expansion

Power Supply Design

The system operates using a standard 24 VAC HVAC supply.

Power Conversion Stages

1. AC to DC Rectification
2. Filtering Stage
3. DC-DC Voltage Conversion
4. LDO Regulation

Generated Voltage Rails

- 34V DC
- 10V DC
- 5V DC
- 3.3V DC

This design ensures stable operation for both power electronics and sensitive digital circuitry.

Communication System

RS-485 Communication

The system uses RS-485 for reliable industrial communication.

Advantages

- Long-distance communication
- Noise immunity
- Multi-drop network capability
- Stable data transmission in HVAC environments

Modbus

Modbus protocol is implemented over RS-485 for structured communication.

Supported Operations

- Status monitoring
- Parameter configuration
- Zone control commands
- Fault reporting

Wi-Fi Monitoring

An ESP32 module enables wireless monitoring and future IoT integration.

Capabilities

- Wireless status updates
- Remote diagnostics

- Future cloud connectivity
- Real-time monitoring

Communication between STM32 and ESP32 is implemented using UART.

Firmware Architecture

The firmware follows a layered architecture for maintainability and scalability.

Firmware Layers

HAL Layer

- GPIO drivers
- UART drivers
- I2C drivers
- Timer interfaces

Communication Layer

- Modbus handling
- UART communication
- Data buffering
- CRC verification

Control Logic Layer

- Relay control
- Damper control logic
- Safety checks
- Limit switch handling

Application Layer

- System initialization
- Task scheduling
- Fault management
- System coordination

Control Algorithm

5.2 Controller Board Design

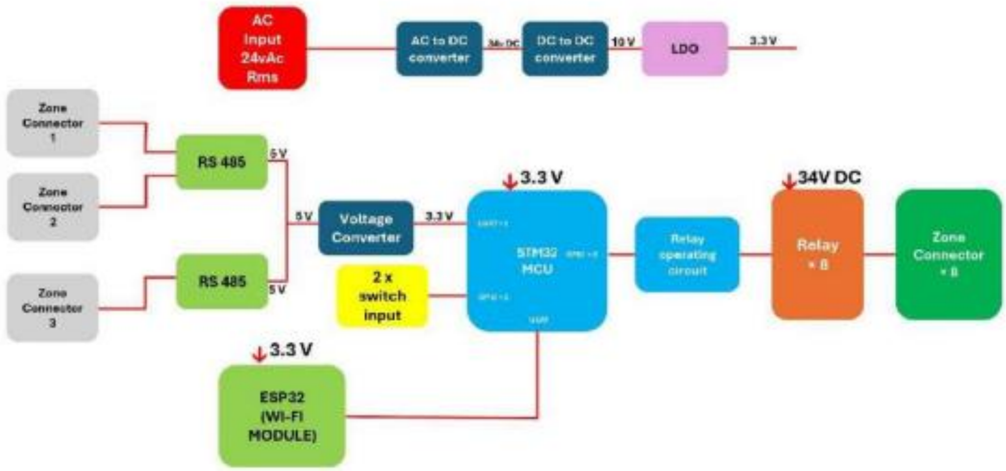


Figure 5.2: Controller Board Block Diagram

5.3 Peripheral Board Design

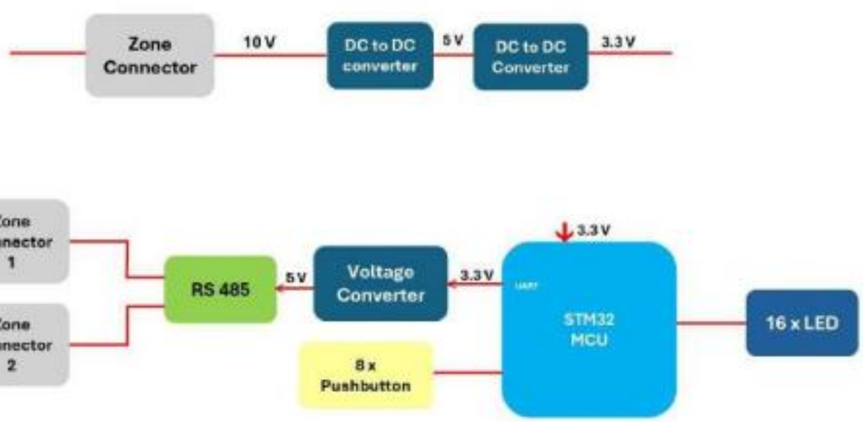


Figure 5.3: Peripheral Board Block Diagram

5.4 AC Motor Board Design

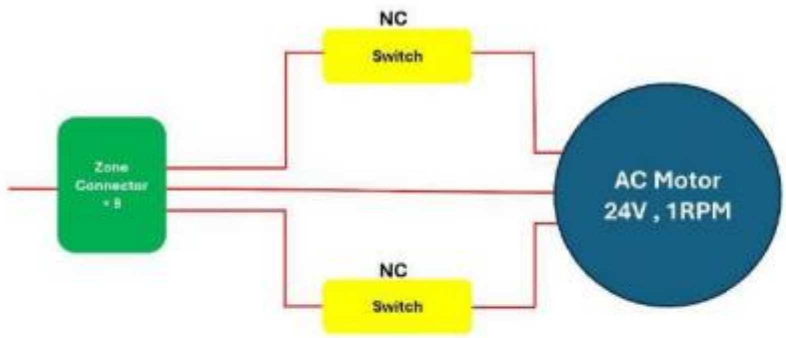


Figure 5.4: AC Motor Board Block Diagram

System Workflow

1. System Initialization
2. Input Reading
3. Decision Making
4. Relay & Motor Control
5. Safety Monitoring
6. Continuous Real-Time Loop

Safety Features

- Limit switch enforcement
- Communication timeout detection
- Fail-safe motor shutdown
- Fault-state handling

Testing & Validation

The system was validated using multiple testing stages.

7.2 Test Cases

Test Case ID	Test Description	Input / Condition	Expected Result	Actual Result	Status
TC01	Power supply verification	24 VAC applied	Stable DC outputs generated	Stable outputs observed	Pass
TC02	STM32 MCU startup	Power ON	MCU initializes correctly	Successful initialization	Pass
TC03	RS-485 communication	Data request sent	Correct data received	Data received correctly	Pass
TC04	Peripheral board input	Pushbutton pressed	Input detected by controller	Input detected	Pass
TC05	LED status indication	System active	LEDs indicate correct status	LEDs functioning	Pass
TC06	Relay operation	Control command issued	Relay switches ON/OFF	Relay switching correct	Pass
TC07	AC motor operation	Zone command issued	Motor runs at 1 RPM	Motor operates smoothly	Pass
TC08	Limit switch detection	End position reached	Motor stops immediately	Motor stopped	Pass
TC09	ESP32 Wi-Fi connection	Wi-Fi enabled	Connection established	Connected successfully	Pass
TC10	System fault handling	Communication timeout	System enters safe state	Safe state achieved	Pass

Testing Performed

- Power supply verification
- STM32 startup testing
- RS-485 communication testing
- Relay operation testing
- AC motor testing
- Limit switch testing
- Wi-Fi communication testing
- Fault handling validation

Result Summary

- Stable power operation
- Reliable RS-485 communication
- Accurate relay switching
- Smooth AC motor control
- Safe limit switch operation
- Stable Wi-Fi connectivity
- Successful fault handling

Advantages

- Modular and scalable design
- Industrial-grade communication reliability
- Embedded real-time control
- Easy maintenance and expansion
- Improved zone-wise airflow management
- Reduced energy wastage
- Future IoT integration support

Future Improvements

- Temperature and humidity sensor integration
- Cloud dashboard development
- Mobile application support
- Advanced analytics
- BACnet / Modbus TCP support
- Predictive control algorithms
- Intelligent energy optimization

Technologies Used

Hardware

- STM32 Microcontroller
- ESP32 Wi-Fi Module
- RS-485 Transceiver
- Relay Driver Circuits
- AC Damper Motors
- Limit Switches

Software & Firmware

- Embedded C
- STM32CubeMX
- HAL Drivers
- UART Communication
- Modbus

Applications

- Smart Building Automation
- HVAC Zone Control
- Industrial Airflow Systems
- Energy-Efficient Building Systems
- Remote Monitoring Solutions

Conclusion

The HVAC Controller System demonstrates a reliable and scalable embedded solution for modern HVAC automation. The project combines modular hardware, structured firmware, industrial communication protocols, and wireless monitoring into a future-ready architecture suitable for smart building environments.

The system successfully achieves:

- Reliable zone control
- Stable communication
- Safe motor operation
- Modular scalability
- Embedded real-time performance

This project provides a strong foundation for future HVAC automation and IoT-based building management systems.